

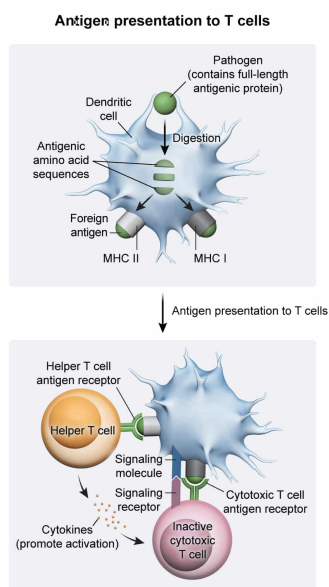
Which series shows the order by which an antigenic sequence of amino acids is brought in contact with a T-cell receptor by a major histocompatibility complex (MHC)?

- ☐ A. Full-length protein → MHC I → MHC II → T-cell receptor [10%]
- ☐ B. Full-length protein → MHC II → MHC I → T-cell receptor [9%]
- ✓ ☒ C. Full-length protein → free digested protein fragments → MHC II → T-cell receptor [66%]
- ☐ D. Full-length protein → free digested protein fragments → MHC I → free digested protein fragments → T-cell receptor [13%]

Correct

66% answered correctly

Explanation:



Because choice C specifies MHC II, the correct sequence would be: full-length protein → free digested protein fragments → MHC II → (helper) T-cell receptor.

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The **immune system** is capable of mounting both "nonspecific" (**innate immunity**) and "specific" (**adaptive** or acquired immunity) responses as part of its defense against invading organisms or particles and abnormal endogenous cells (eg, cancer cells). Although components of both innate and adaptive immunity recognize specific abnormalities in the cells or particles they target, the adaptive immune system is very narrowly targeted (ie, "specific") whereas the focus of the innate immune system is much broader (hence, "nonspecific").

The adaptive immune system consists of **humoral immunity**, mediated by **B cells**, and **cell-mediated immunity**, mediated by T cells. T cells are only activated by antigen presentation.

When **antigens are presented** on the surface of cells, they are in complex with major histocompatibility complex (MHC) proteins. All nucleated cells have MHC I molecules, but certain immune cells (eg, **macrophages**, lymphocytes, dendritic cells) also contain MHC II molecules and are known as "professional" antigen-presenting cells.

The question asks for the order by which an antigenic (ie, producing an immune response) sequence of amino acids is brought in contact with a T-cell receptor by an MHC. Infected cells **break down pathogen molecules** (ie, bacterial or viral) **into smaller fragments**. Some of

these fragments can form **complexes with MHC I or II molecules** that are **recognized** (ie, bound) **by T-cell receptors**. Therefore, the best series order for this pathway is **full-length protein → free digested protein fragments → MHC II → T-cell receptor**.

(Choices A and B) Antigenic digested protein fragments can be transferred to either MHC I or II, but such fragments are not transferred from one MHC to another.

(Choice D) When antigenic protein fragments are transferred to MHC I, they are presented to the T cell as a complex with MHC I, not as free fragments newly dissociated from MHC I.

Educational objective:

All nucleated cells contain major histocompatibility complex (MHC) I molecules, and "professional" antigen-presenting cells also contain MHC II molecules. MHC molecules form complexes with antigens that are displayed on the surface of cells. Antigens displayed in complex with MHC molecules on the surface of cells bind to T-cell receptors.

Time spent:0 sec

Subject:
Biology

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System:
Skin and Immune Systems